Practical PCB Design

ECEN 3730

Board 2: Good vs Bad Layout

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Project Overview:

This board was made to demonstrate the importance of certain best design practices. There were two identical copies of a hex inverter circuit placed on the same board, one routed using good practices (i.e. close decoupling capacitor, short path lengths, continuous ground plane), and one routed with bad practices. This board also incorporated an LDO to allow either 5V or 3.3V power to the IC V_{cc} pins.

Circuit Plan and Theory:

Function

Functionality of this board should include

- Switching between 5V input from barrel jack and 3.3V LDO output
- 555 timer circuit configured in astable mode, outputting 500Hz, 50% duty cycle square wave
- Two identical hex inverter circuits, one designed with best practices, one designed with bad practice
    - Hex circuits output to 3 LEDs, and quiet high and quiet low test points

My initial block diagram for this board is shown in Figure 1.

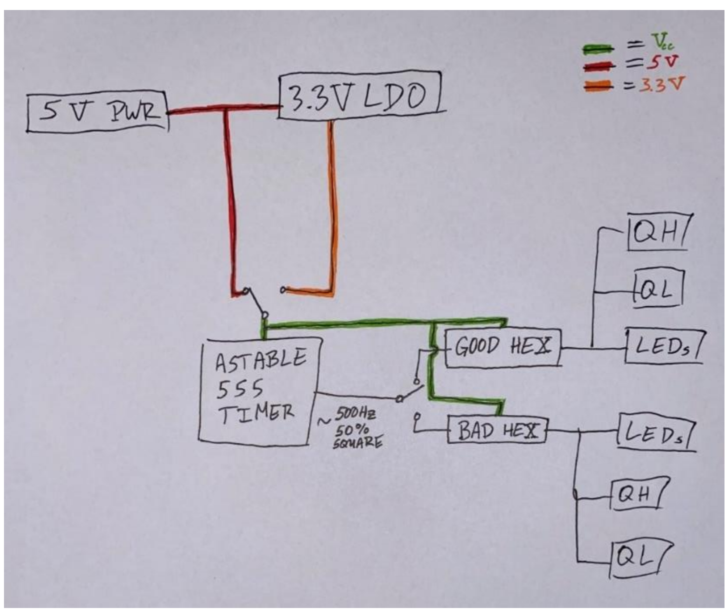


Fig. 1 (Functional block diagram for Board 2)

Schematic

Figure 2 below shows the fully fleshed-out schematic for this board based on the block diagram in Figure 1 above.

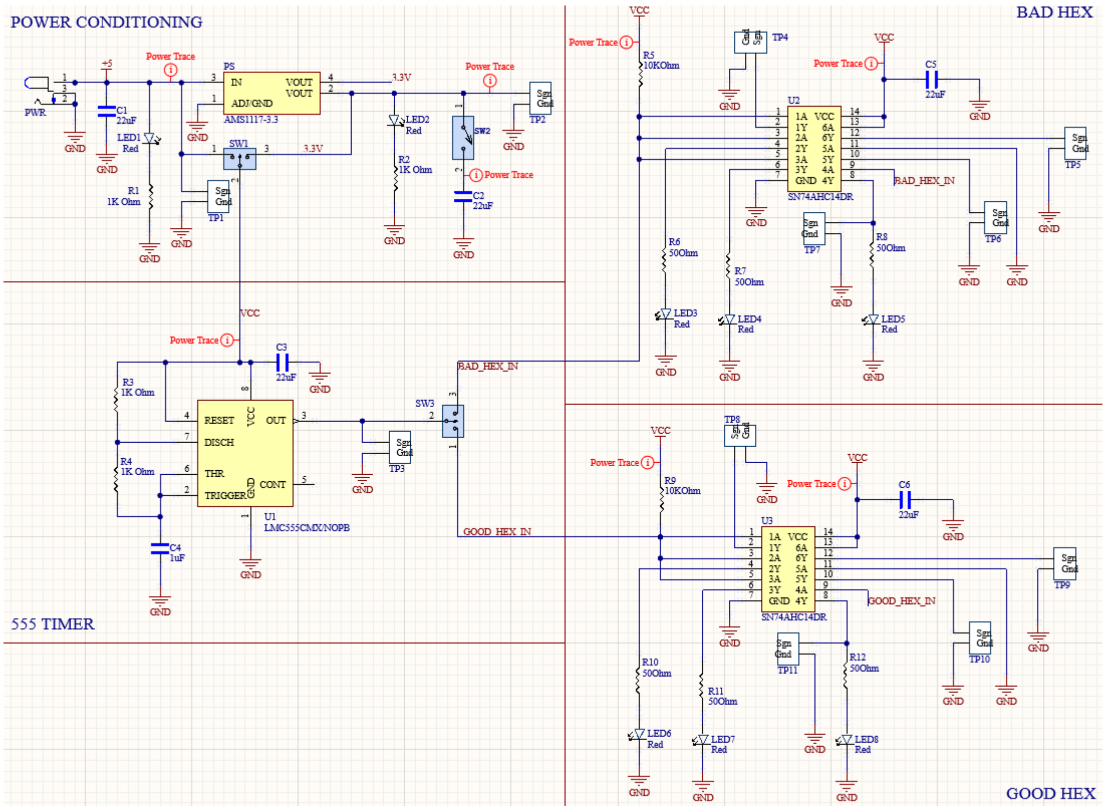


Fig. 2 (Schematic for Board 2)

- Timer circuit designed with fast 555 timer
    - TA mentioned in class that slow might not operate at 5V Vcc
    - Faster switching should cause more noise at outputs, will be easier to compare
- 3-way switch at timer output to operate good and bad hex independently
- 3-way switch for 5V power and 3.3V LDO
- Test points for 5V power, 3.3V LDO power, timer output, and hex inverter outputs

PCB Layout

Figure 3 below shows the layout done in Altium of this board.

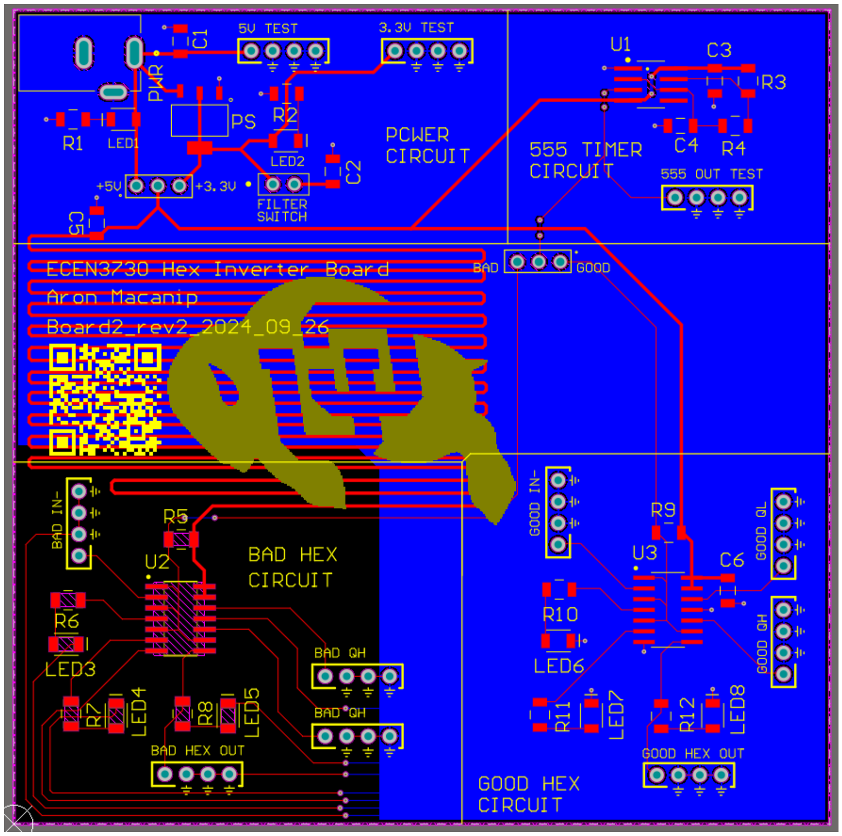
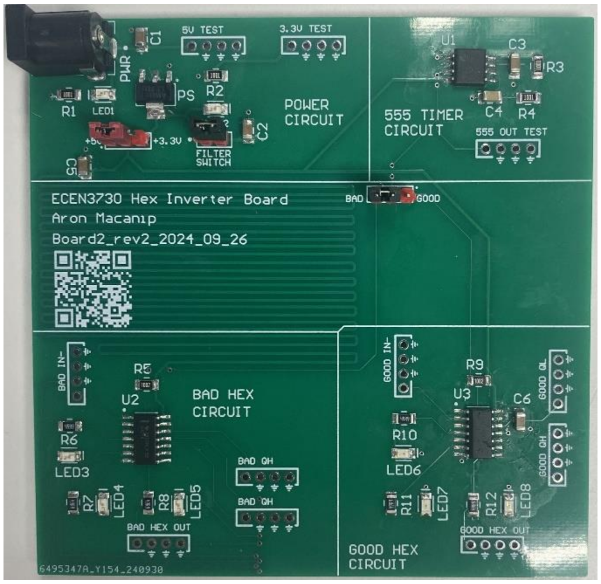


Fig. 3 (Layout of Board 2)

- Long path length to bad hex to increase inductance for max switching noise to bad hex
- 3.9”x3.9” board size to maximize path length from bad hex to its decoupling capacitor within standard JLC board size constraint
- Good and band hex components laid out as identically as possible for more directly comparison between good and bad practice
- Silk screen labels at switches for quick determination of operating mode
- Subcircuits partitioned and labeled on silk screen layer for clear visual of which components serve which function

Physical Board:

Figures 4a and 4b below show the fully assembled physcial board.



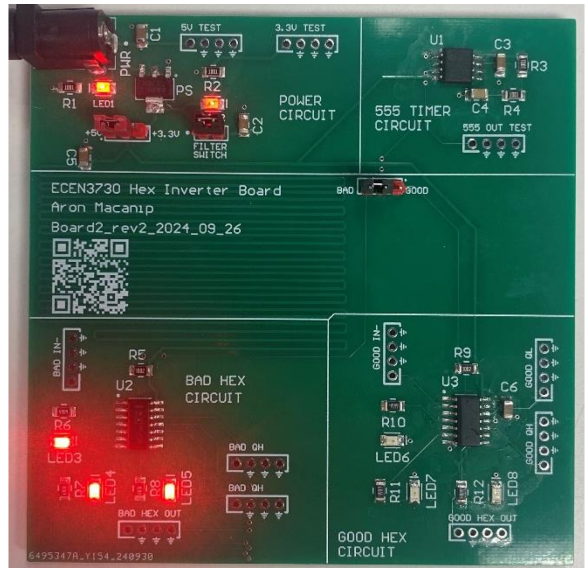


Fig. 4a (Fully assembled Board 2) Fig. 4b (Board 2 powered on)

- All test points routed correctly
- Partially delaminated pin 13 of good hex IC while soldering
    - Quiet low pin meant to be tied to V_{CC}
    - Worst case scenario = quiet low outputs roughly same as quiet high pin
    - No noticeable effect when measured
- Soldered power circuit first, verified 5V and 3.3V outputs before finishing board
- Had to wait to receive proper 555 timer
    - Used jumpers to make power connection to breadboarded astable 555 timer
    - Connected breadboarded 555 output to hex circuits using jumpers to 3-way header to verify functionality while waiting for SMT timer
- 5V and 3.3V LEDs not labeled, bad quiet low mislabeled as quiet high

Measurements

Scope key:

- Yellow trace -> V_{CC} measurement
- Blue trace -> 555 timer output/trigger
- Green trace/orange reference -> Good hex measurement
- Red trace -> Bad hex measurement

LDO

Figures 5 and 6 below show the difference between the filtered and unfiltered LDO output. Both were triggered on the rising edge of 555 timer.

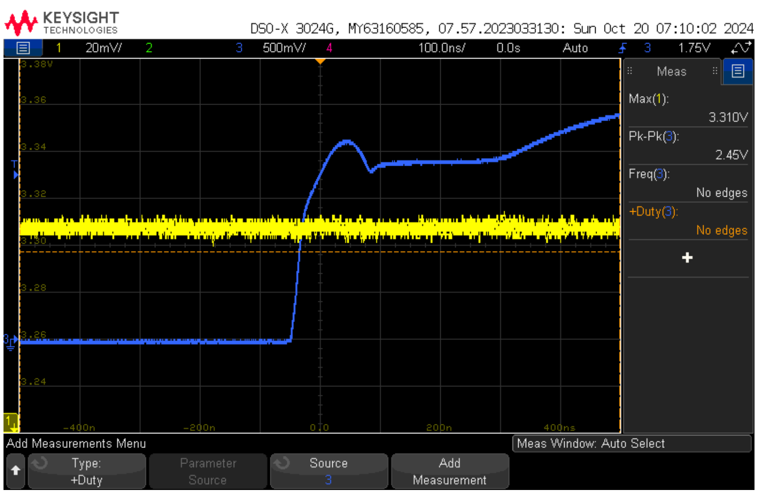


Fig. 5 (LDO w/ filter capacitor)

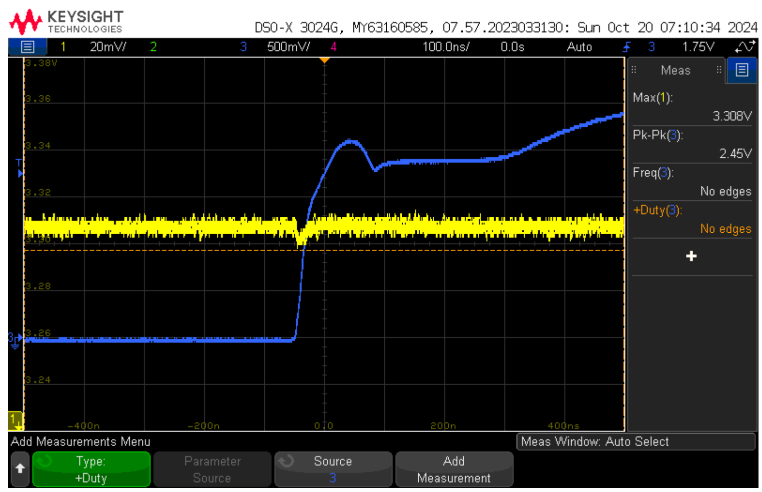


Fig. 6 (LDO w/o filter capacitor)

Without a filter capacitor, the LDO output has ~10mV drop on the rising edge of timer due to fast current switching.

Good Hex Inverter

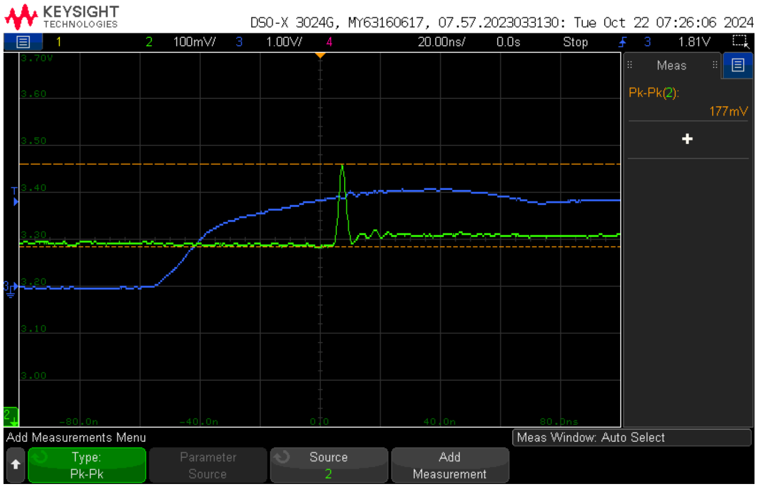


Fig. 7a (3.3V quiet high rise noise)

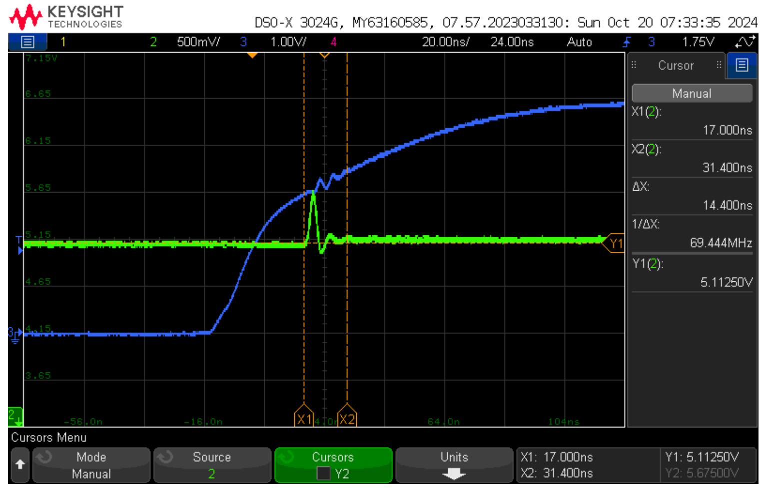


Fig. 7b (5V quiet high rise noise)

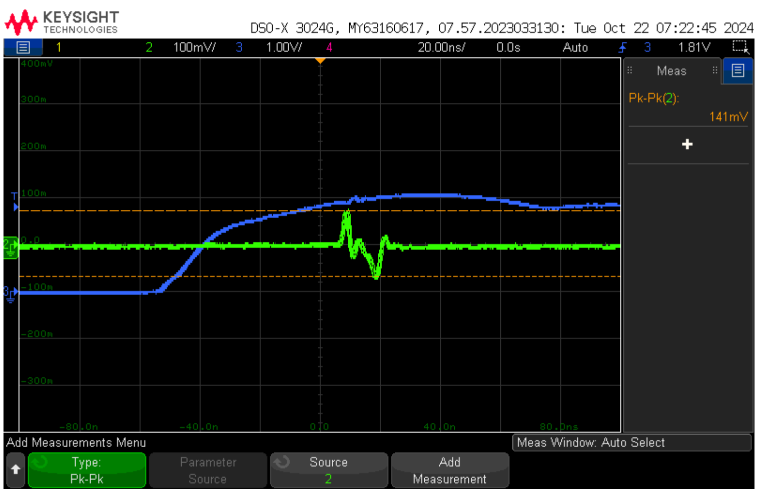


Fig. 8a (3.3V quiet low rise noise)

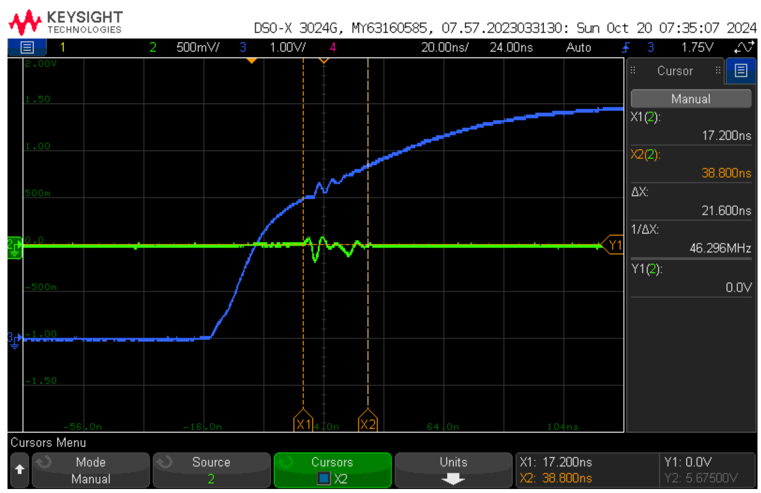


Fig. 8b (5V quiet low rise noise)

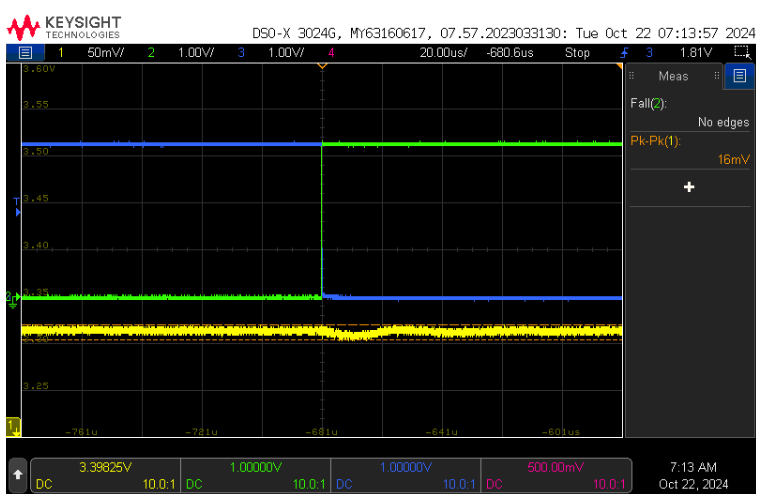


Fig. 9 (3.3V power rail rise noise)

Bad Hex Inverter

Orange reference scope traces are measurements of the same parameter from the good hex layout and are references for scale.

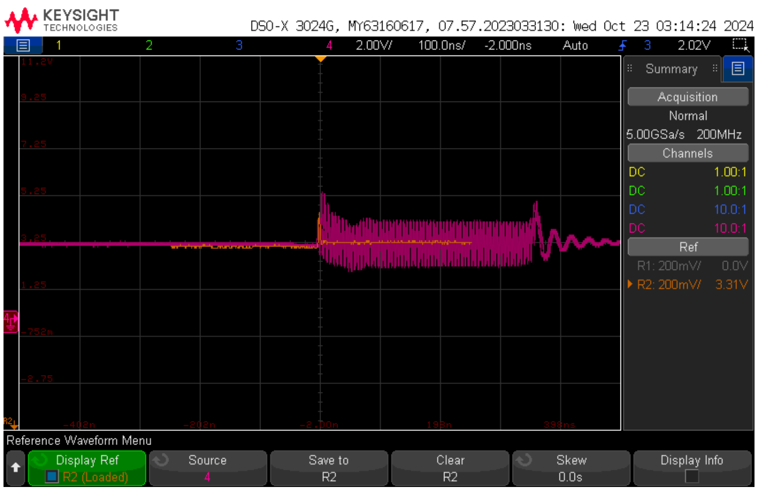


Fig. 10a (3.3V quiet high noise, 10:1 scale)

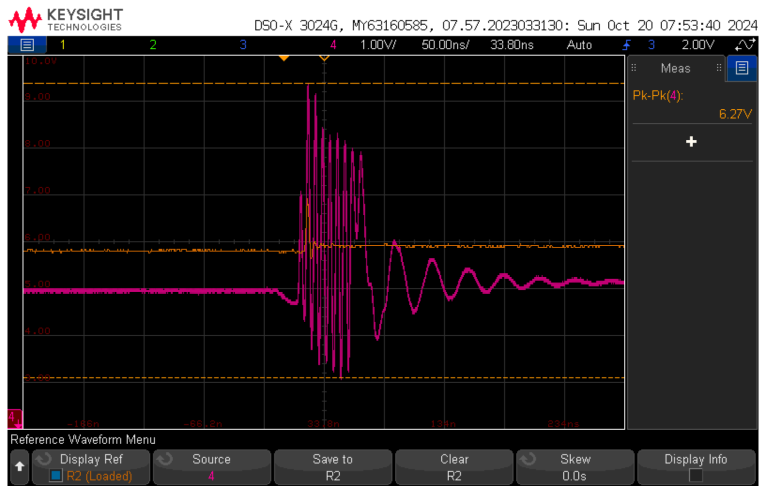


Fig. 10b (5V quiet high noise, 2:1 scale)

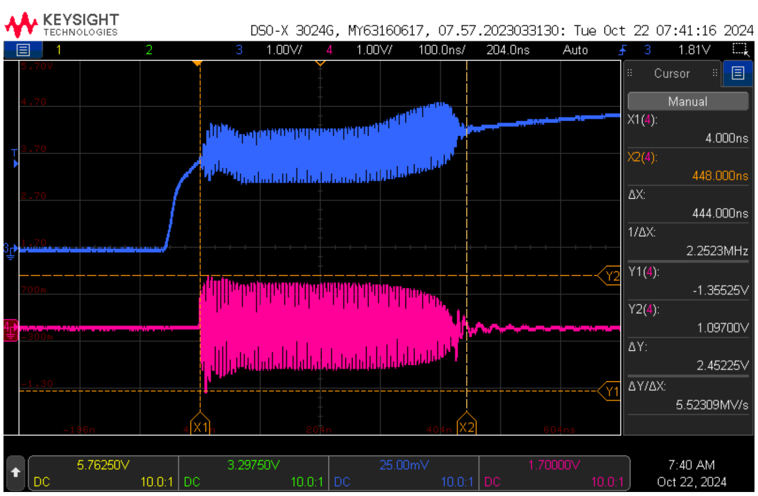


Fig. 11a (3.3V quiet low noise)

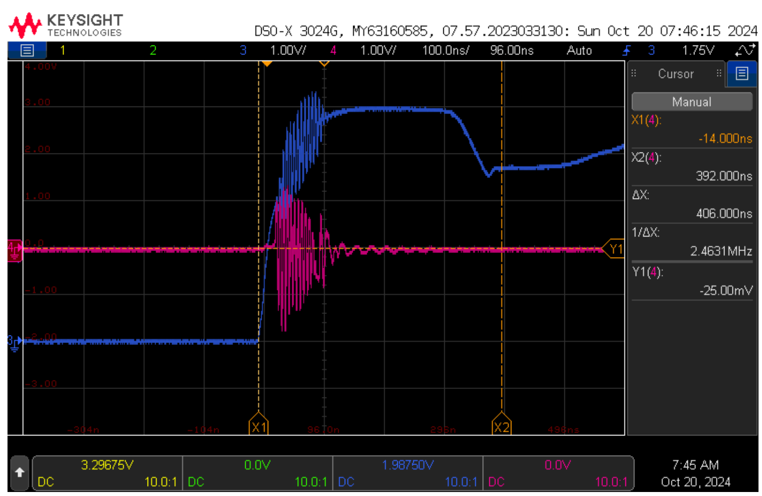


Fig. 11b (5V quiet low noise)

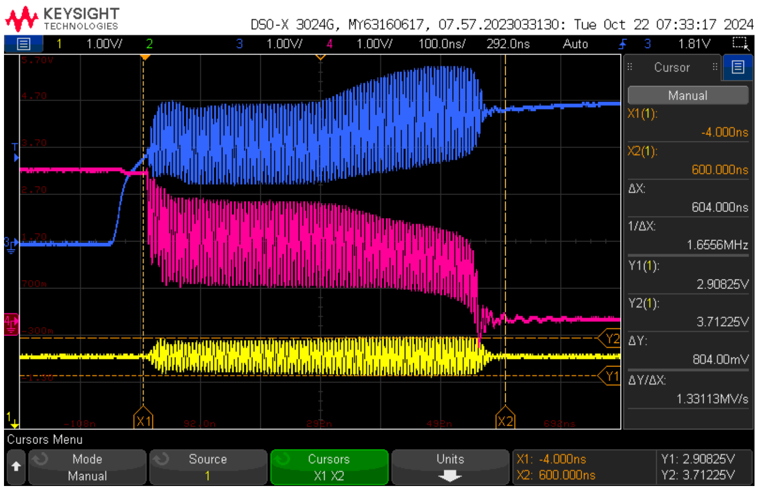


Fig. 12 (3.3V power rail noise)

Summary

	V_{CC}		Voltage	Duration
Good Hex	3.3V	3.3V Quiet High	177 mV Pk-Pk	10 ns
		3.3V Quiet Low	141 mV Pk-Pk	15 ns
		Power Rail Noise	16 mV Pk-Pk	60 µs
	5V	5V Quiet High	100 mV Pk-Pk	14 ns
		5V Quiet Low	400 mV Pk-Pk	22 ns
Bad Hex	3.3V	3.3V Quiet High	3 V Pk-Pk	500 ns
		3.3V Quiet Low	2.45 V Pk-Pk	600 ns
		Power Rail Noise	804 mV Pk-Pk	600 ns
	5V	5V Quiet High	6.3 V Pk-Pk	250 ns
		5V Quiet Low	3 V Pk-Pk	250 ns

Table 1 (Summary of good and bad hex measurements)

- Significant peak-to-peak voltage swing on bad hex
  - Farther decoupling capacitor, very large loop inductance cause large voltage swings at IC power
  - > 10x noise duration means much higher currents through bad hex

Areas for Potential Improvement

- Make sure silk screen labels are correct
- Place good hex layout closer to 555 timer output to minimize signal and power path lengths
- Make bad hex layout less bad
  - Bad layout was so bad it was difficult to point out subtler differences and draw more meaningful conclusions

Conclusions:

- Short paths between power pins and associated decoupling capacitors minimize loop inductance
  - Less switching noise to power pins
- Filter capacitor with LDO can reduce potential noise at output due to feedback oscillations
- Keep ground connections as short and direct as possible
  - Continuous ground plane for short return paths
- Be mindful of heat applied during soldering
  - Not too much heat at once, allow pads to cool if struggling to get good solder joint